2. Optimization (stabilizing and scaling)

This document focuses on optimizing and stripping out what's already up and running without breaking the base and the bank.

> This optimization phase focuses on eliminating inefficiencies and aligning infrastructure with workload characteristics, ensuring cost-effective, high-performance operations without disrupting existing development velocity.

This phase focuses on improving the current infrastructure by:

- Reducing **cloud expenditure**
- Eliminating **inefficiencies**
- Aligning workloads with **appropriate environments (cloud vs on-prem)**
- Introducing **non-intrusive DevSecOps practices**
- Preparing for **controlled modernization**.

Objective is to reinvest these cost saving to take these workloads to a HA.

With all this, the expect outcome to achieve
1) Cost
- Reduction of **~30–40% in cloud spend**
- Elimination of ide/waste resources
- Achieve Savings up to **~$8.5K+ USD/month (~₹6–7 Lakhs)** in cloud bills.
2) Performance
- Improved latency (on-prem media handling)
- Reduced network bottlenecks and cross-region traffic
3) Security
- Increased visibility into vulnerabilities
- Gradual DevSecOps adoption
4) Operations
- Retain existing high-speed deployment workflows
- Introduce **parallel feedback pipelines**
- Improve system observability and operational control

Problem statements:

Fixing:

1. **Infrastructure Footprint**
    - Scattered Resources

- No regional strategy
- High networking complexity
- Cost leakage via unused / scattered infra

2. **Operational Gaps**
   - No Infrastructure as Code (manual provisioning)
   - No standardized deployment model
   - Resource ownership unclear → stale resources
   - No audit / lifecycle governance
   - CI/CD exists but:
     - No SecOps
     - No feedback loops
3. **Cost & Performance Pressure**

- ~42% cost increase trend
- Major contributors:
  - EC2 (compute-heavy workloads)
  - RDS (persistent usage)
  - VPC/network transfer (voip traffic)

Action Items

A. Eliminating Idle & Stale Resources

- [ ] Identify and remove:
  - [ ] Unused EC2 instances
  - [ ] Detached EBS volumes
  - [ ] Orphaned load balancers
  - [ ] Unused Elastic IPs and snapshots

Resource Elimination (Action: Stop/Delete idle or unused resources

Resource Type: EBS volume
Recommended resource summary: Detach volume from instance, create a snapshot and delete. / Create a snapshot and delete.
Estimated savings percentage: 100%
Implementation Effort: Low
resource restart needed: No

rollback possible: No
Account name and ID: Pulse Telesystems Private Limited (647096474516)

Resource ID	Top recommended action	Estimated monthly cost savings	Account name and ID	Region
vol-084e5af34c088b066	Delete idle or unused resources	$0.64	Pulse Telesystems Private Limited (647096474516)	US East (Ohio)
vol-0446ea77ae6c7e383	Delete idle or unused resources	$0.64	Pulse Telesystems Private Limited (647096474516)	US East (Ohio)
vol-0c48cc4b79da98950	Delete idle or unused resources	$0.73	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-03f2c92500836c463	Delete idle or unused resources	$0.73	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-028d577dd2db989cc	Delete idle or unused resources	$0.73	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0cbbb839e2b7003ba	Delete idle or unused resources	$0.73	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0ae003ee621802381	Delete idle or unused resources	$0.73	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-03e98972ce2d862d1	Delete idle or unused resources	$0.91	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-058242a64377a1ed7	Delete idle or unused resources	$0.91	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0bd98f907b00807c0	Delete idle or unused resources	$0.91	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-002e439a76fcd189d	Delete idle or unused resources	$0.91	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)

Resource ID	Top recommended action	Estimated monthly cost savings	Account name and ID	Region
vol-0f7df439d195169de	Delete idle or unused resources	$0.91	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0137d8b095862c68c	Delete idle or unused resources	$0.91	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-00441e0dea91e6ed4	Delete idle or unused resources	$0.96	Pulse Telesystems Private Limited (647096474516)	US West (N. California)
vol-0a01db26260fa7b48	Delete idle or unused resources	$1.20	Pulse Telesystems Private Limited (647096474516)	US East (Ohio)
vol-006a7e7aba9bd85c7	Delete idle or unused resources	$1.20	Pulse Telesystems Private Limited (647096474516)	US East (Ohio)
vol-06bb0289d63a08f10	Delete idle or unused resources	$1.37	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-051797d2fb6ac4bd4	Delete idle or unused resources	$1.37	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0e102428452a7bc2e	Delete idle or unused resources	$1.37	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-08389d995339c0dfe	Delete idle or unused resources	$1.37	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
Resource ID	**Top recommended action**	**Estimated monthly cost**	**Account name and ID**	**Region**
vol-0a7fc49e813659b42	Delete idle or unused resources	$1.37	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-07f5d520b11c735b4	Delete idle or unused resources	$1.37	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)

Resource ID	Top recommended action	Estimated monthly cost savings	Account name and ID	Region
vol-00f5eaa0a9465c774	Delete idle or unused resources	$1.50	Pulse Telesystems Private Limited (647096474516)	US East (Ohio)
vol-01436cd5d60fb2f73	Delete idle or unused resources	$1.60	Pulse Telesystems Private Limited (647096474516)	US East (Ohio)
vol-0405c7bd047186113	Delete idle or unused resources	$1.60	Pulse Telesystems Private Limited (647096474516)	US East (Ohio)
vol-029d70ca87a7c9e72	Delete idle or unused resources	$1.82	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0f372853a9c66883e	Delete idle or unused resources	$1.82	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-074c42a36c65191ad	Delete idle or unused resources	$1.82	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0c0a96549fda2389e	Delete idle or unused resources	$1.82	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-05732d9a8b8fc1a90	Delete idle or unused resources	$1.82	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0fe36933983c1eace	Delete idle or unused resources	$1.82	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-00c20b556676b9ca4	Delete idle or unused resources	$1.82	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-08317ea3615fcfbee	Delete idle or unused resources	$1.82	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-089636e58c70a4e35	Delete idle or unused resources	$1.82	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)

Resource ID	Top recommended action	Estimated monthly cost savings	Account name and ID	Region
vol-0469e52eefe2d1871	Delete idle or unused resources	$2.28	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-093fbec1887c0fed4	Delete idle or unused resources	$2.28	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-08e1e82963034cb7a	Delete idle or unused resources	$2.28	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-046f652cdbfa5a222	Delete idle or unused resources	$2.28	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-04da5f5e4e014e714	Delete idle or unused resources	$2.40	Pulse Telesystems Private Limited (647096474516)	US East (Ohio)
vol-0b8f0b8f6b053e432	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
Resource ID	**Top recommended action**	**Estimated monthly cost**	**Account name and ID**	**Region**
vol-0438263ebb005003c	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-08059674b09053693	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0fe7d6903d9c1388f	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-09b760682c0dacd67	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0e92b2c93a4f8e9e2	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)

Resource ID	Top recommended action	Estimated monthly cost savings	Account name and ID	Region
vol-08c5e0184066d6595	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-081a4634314933a53	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0cb6dc75e9b9f9702	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-03e73a282881fca29	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-018763d02a3bcb094	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-03676d0f253615f95	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-05da2bea83edd29c0	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-02bc5bbec1afd7944	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-03f7289efdf9f756e	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0fee25414e50f9cc0	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0bd2351cd5a38a401	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-02f3e329b331bb59d	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)

Resource ID	Top recommended action	Estimated monthly cost savings	Account name and ID	Region
vol-09b940f29eefffe2e	Delete idle or unused resources	$2.74	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-06045601624d435b5	Delete idle or unused resources	$3.30	Pulse Telesystems Private Limited (647096474516)	Canada (Central)
vol-08b56a849ba8a02c7	Delete idle or unused resources	$3.30	Pulse Telesystems Private Limited (647096474516)	Canada (Central)
Resource ID	**Top recommended action**	**Estimated monthly cost**	**Account name and ID**	**Region**
vol-0d8e0012a54e1709d	Delete idle or unused resources	$3.30	Pulse Telesystems Private Limited (647096474516)	Canada (Central)
vol-08f5742c01e4ab70e	Delete idle or unused resources	$3.42	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0cf8da9bb0512f3fe	Delete idle or unused resources	$4.00	Pulse Telesystems Private Limited (647096474516)	US East (Ohio)
vol-0fa44cdc3750fa4fb	Delete idle or unused resources	$4.00	Pulse Telesystems Private Limited (647096474516)	US East (Ohio)
vol-0d42a1e438587f1e4	Delete idle or unused resources	$4.00	Pulse Telesystems Private Limited (647096474516)	US East (Ohio)
vol-0b40bdcc43d264184	Delete idle or unused resources	$4.56	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0323a607389c2cda4	Delete idle or unused resources	$4.56	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0af56f46b1b49b904	Delete idle or unused resources	$4.56	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)

Resource ID	Top recommended action	Estimated monthly cost savings	Account name and ID	Region
vol-0bc6b8f08a5d67021	Delete idle or unused resources	$4.56	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-09a908d6a86179c92	Delete idle or unused resources	$4.56	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0e9b609e5f8e71ea7	Delete idle or unused resources	$5.02	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0b708a046f7518521	Delete idle or unused resources	$5.47	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-02d8299c301785c94	Delete idle or unused resources	$5.47	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-06aba7a65c3995673	Delete idle or unused resources	$5.47	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0caa4b2d3314031b2	Delete idle or unused resources	$5.70	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0b5975a64e5aae2ef	Delete idle or unused resources	$6.84	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0725a21bbf853cdc5	Delete idle or unused resources	$6.84	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0471bfadfdc117960	Delete idle or unused resources	$8.21	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)
vol-0d613aca3600a0d0f	Delete idle or unused resources	$15.96	Pulse Telesystems Private Limited (647096474516)	Asia Pacific (Mumbai)

B. Right-Sizing Resources:

- [ ] Downscale:
  - [ ] Underutilized EC2 instances
  - [ ] Over-provisioned RDS DBs
  - [ ] Introduce Spot Instances/Batch for schedule jobs and workloads
- [ ] Introduce:
  - [ ] Auto-scaling wherever applies
  - [ ] Workload based scaling

Also refer: AWS Compute Optimizer

Resource Type: EC2 Instance
Top recommended action: Rightsizing - EC2
Estimated savings percentage: 100%
Implementation Effort: Medium, High
resource restart needed: Yes
rollback possible: Yes
Estimated savings percentage: 1% - 56%
Account name and ID: Pulse Telesystems Private Limited (647096474516)

Instance Rightsizing List: (Action: Rightsizing)

Estimated monthly savings	Resource ID	Current resource summary	Recommended resource summary	Estimated monthly cost	Region
$0.33	i-0cbfbdd159517b34e	m5a.large	m6a.large	$40.88	Asia Pacific (Mumbai
$4.90	i-006640fcd241e234d	m5a.large	t3a.large	$40.88	Asia Pacific (Mumbai
$4.90	i-0788a191c99e16379	m5a.large	t3a.large	$40.88	Asia Pacific (Mumbai
$4.90	i-02162c374a51bd6a4	m5a.large	t3a.large	$40.88	Asia Pacific (Mumbai
$4.90	i-08a2da6ff9ab8f650	m5a.large	t3a.large	$40.88	Asia Pacific (Mumbai

Estimated monthly savings	Resource ID	Current resource summary	Recommended resource summary	Estimated monthly cost	Region
$4.90	i-0757e582e054daef5	m5a.large	t3a.large	$40.88	Asia Pacific (Mumbai
$4.90	i-09fa29369cd558d31	m5a.large	t3a.large	$40.88	Asia Pacific (Mumbai
$4.90	i-0b62075f4a123889e	m5a.large	t3a.large	$40.88	Asia Pacific (Mumbai
$4.90	i-0d74578a7bbb89c3d	m5a.large	t3a.large	$40.88	Asia Pacific (Mumbai
$4.90	i-04ba9cd7bd60183ef	m5a.large	t3a.large	$40.88	Asia Pacific (Mumbai
$4.90	i-06d025dcce6683cb0	m5a.large	t3a.large	$40.88	Asia Pacific (Mumbai
$9.06	i-05d2fcd3d1fd197bb	m5a.xlarge	t3a.xlarge	$81.03	Asia Pacific (Mumbai
$18.04	i-0b4bc4ac9c2e17e67	t3a.large	t3a.medium	$35.99	Asia Pacific (Mumbai
$19.70	i-0acab63aa54dfa350	t3a.xlarge	r6a.large	$71.67	Asia Pacific (Mumbai
$19.78	i-03dac8a157ade79ad	t3a.xlarge	r6a.large	$71.94	Asia Pacific (Mumbai
$19.78	i-04a0cae1e5aebe4a8	t3a.xlarge	r6a.large	$71.97	Asia Pacific (Mumbai
$19.79	i-0d434298e66301c12	t3a.xlarge	r6a.large	$71.98	Asia Pacific (Mumbai
$19.79	i-01cef61b3babbaf9e	t3a.xlarge	r6a.large	$71.98	Asia Pacific

Estimated monthly savings	Resource ID	Current resource summary	Recommended resource summary	Estimated monthly cost	Region
					(Mumbai
$19.79	i-00b5f493f0c87dd00	t3a.xlarge	r6a.large	$71.98	Asia Pacific (Mumbai
$19.79	i-079e573f40d322a65	t3a.xlarge	r6a.large	$71.98	Asia Pacific (Mumbai
Estimated monthly savings	**Resource ID**	**Current resource summary**	**Recommended resource summary**	**Estimated monthly cost**	**Region**
$19.79	i-08421cb5eaeb36267	t3a.xlarge	r6a.large	$71.98	Asia Pacific (Mumbai
$19.79	i-0ddbd3ad81c7597a2	t3a.xlarge	r6a.large	$71.98	Asia Pacific (Mumbai
$19.79	i-04ca6598fdc7ad562	t3a.xlarge	r6a.large	$71.98	Asia Pacific (Mumbai
$19.79	i-002fc65be4372e132	t3a.xlarge	r6a.large	$71.98	Asia Pacific (Mumbai
$28.84	i-006b69793a9e132e1	m5a.xlarge	r6a.large	$81.03	Asia Pacific (Mumbai
$28.84	i-066384dd20c282595	m5a.xlarge	r6a.large	$81.03	Asia Pacific (Mumbai
$28.84	i-00b211195575d923e	m5a.xlarge	r6a.large	$81.03	Asia Pacific (Mumbai
$28.84	i-01c7b7a41c4fb469b	m5a.xlarge	r6a.large	$81.03	Asia Pacific (Mumbai
$28.84	i-0c0a83bba84ae5d2b	m5a.xlarge	r6a.large	$81.03	Asia Pacific (Mumbai

Estimated monthly savings	Resource ID	Current resource summary	Recommended resource summary	Estimated monthly cost	Region
$28.84	i-0b5949cf82a33dc5d	m5a.xlarge	r6a.large	$81.03	Asia Pacific (Mumbai
$28.84	i-0a8934f176571303a	m5a.xlarge	r6a.large	$81.03	Asia Pacific (Mumbai
$28.84	i-0ee350c3cfa1d5d2e	m5a.xlarge	r6a.large	$81.03	Asia Pacific (Mumbai
$28.84	i-03bc089cca6bcb771	m5a.xlarge	r6a.large	$81.03	Asia Pacific (Mumbai
$42.11	i-04fd93e0bca1cf07f	t2.xlarge	r6i.large	$122.12	Asia Pacific (Mumbai
$43.07	i-0ec81345372775d4b	m5a.xlarge	r5a.large	$125.55	US East (Ohio)
$43.07	i-068eacac7ed0984ae	m5a.xlarge	r5a.large	$125.56	US East (Ohio)
$57.59	i-056f3fb3259c4b057	m5a.2xlarge	r6a.xlarge	$161.82	Asia Pacific (Mumbai
$57.64	i-033ae54d47001143e	m5a.2xlarge	r6a.xlarge	$161.98	Asia Pacific (Mumbai
$57.65	i-0f49af4ac3f6fe327	m5a.2xlarge	r6a.xlarge	$161.99	Asia Pacific (Mumbai
$57.67	i-0b7375ebf0388f1e8	m5a.2xlarge	r6a.xlarge	$162.04	Asia Pacific (Mumbai
Estimated monthly savings	**Resource ID**	**Current resource summary**	**Recommended resource summary**	**Estimated monthly cost**	**Region**
$57.67	i-0cbbb8ea753f43df4	m5a.2xlarge	r6a.xlarge	$162.05	Asia Pacific (Mumbai

Estimated monthly savings	Resource ID	Current resource summary	Recommended resource summary	Estimated monthly cost	Region
$57.67	i-03554bb7f2ce9d5e6	m5a.2xlarge	r6a.xlarge	$162.06	Asia Pacific (Mumbai
$85.98	i-0aa3ce02fa349a1c1	m5a.2xlarge	r5a.xlarge	$250.62	US East (Ohio)
$115.34	i-09a10d16f980c512a	m5a.4xlarge	r6a.2xlarge	$324.12	Asia Pacific (Mumbai
$278.35	i-04a8bc694b357cb88	c6g.8xlarge	r7g.2xlarge	$497.57	Asia Pacific (Mumbai

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C. Network and Compute Optimization

- [ ] Evaluate:
  - [ ] Evaluate workloads using Migration Accelerator before migration
  - [ ] Migration to ARM-based instances (Graviton) wherever necessary (Instance types with actual workload usage)
  - [ ] Reduction of cross-region / cross-AZ traffic
  - [ ] Optimization of API traffic (GET/POST efficiency)
  - [ ] Establish: Site-to-site VPN OR WireGuard/Tailscale mesh
  - [ ] Low-latency secure communication between: AWS ↔ On-Prem
- [ ] Optimize:
  - [ ] Hybrid connectivity (VPN / Direct Connect tuning)
  - [ ] Data transfer paths (especially RTP/media flows)
- [ ] Review:
  - [ ] Savings Plans / Reserved Instance coverage
- Ref:
  - AWS Documentation
  - Medium article - AWS A simple way to migrate from x86 to ARM
  - VPN Interconnectivity at scale for Hybrid Infrastructure

Saving and Reservation Savings: (Action: Purchase Reservations, Purchase Savings Plans)

Top recommended action: Purchase Reservations
Implementation effort: Very Low
resource restart needed: No
rollback possible: No
Account name and ID: Pulse Telesystems Private Limited (647096474516)

Estimated monthly savings	Resource type	Recommended resource summary	Estimated savings percentage	Estimated monthly cost	Region
US$23.31	ElastiCache Reserved Nodes	4 cache.t4g.micro in ap-south-1 with three years term (AllUpfront) in AWS account 647096474516 (Linked account level)	47%	US$49.66	Asia Pacific (Mumbai)
US$614.83	RDS Reserved Instances	4 db.r5.large Aurora MySQL in ap-south-1 with three years term (AllUpfront) in AWS account 647096474516 (Linked account level)	64%	US$963.60	Asia Pacific (Mumbai)
US$5,491.44	Compute Savings Plans	$6.691/hour with three years term (AllUpfront) in AWS account 647096474516 (Linked account level)	46%	US$11,964.53	-

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D. Storage Optimization

- [ ] Implement:
  - [ ] S3 lifecycle policies (Hot → Cold → Glacier)
  - [ ] Intelligent tiering

- [ ] Remove:
  - [ ] Redundant backups
  - [ ] Unused storage volumes

E. Operational Excellence

- [ ] Enable:
  - [ ] EC2 / EKS Auto Scaling Groups
  - [ ] Kubernetes HPA (Horizontal Pod Autoscaling)
  - [ ] Have `eksctl` on machines to manage cluster
  - [ ] **Karpenter** (dynamic node provisioning)
- [ ] Evaluate:
  - [ ] Migration to cost-efficient service alternatives where applicable
- [ ] Evaluate (ref:) Trusted Advisor (Purchase after consulting with Partner)
  - [ ] Cost optimization
  - [ ] Performance
  - [ ] Security
  - [ ] Fault tolerance
  - [ ] Service limits
  - [ ] Operational excellence

Recommendations - Checks Summary:

Security: Action recommended: 2

- **Amazon S3 Bucket Permissions**: Checks buckets in Amazon Simple Storage Service (Amazon S3) that have open access permissions or allow access to any authenticated AWS user. 44 of 63 buckets have permission properties that grant global access.
- **Security Groups - Specific Ports Unrestricted**: Checks security groups for rules that allow unrestricted access (0.0.0.0/0) to specific ports.158 of 2251 security group rules allow unrestricted access to a specific por

Service Limits: Investigation recommended: 2

- **EC2-VPC Elastic IP Address**: Checks for usage that is more than 80% of the EC2VPC Elastic IP Address Limit. 1 of 16 items have usage that is more than 80% of the service limit.
- **VPC**: Checks for usage that is more than 80% of the VPC Limit. 1 of 16 items have usage that is more than 80% of the service limit.

F. Fence the Regions

- [ ] Reduce from **29 regions → 2–3 strategic regions**
- [ ] Define:
  - [ ] Primary region (production) - Asia/Kolkata
  - [ ] Secondary (DR / failover) - N. Virginia

G. Continuous Audit and Review

- [ ] Leverage:
  - [ ] AWS Cost Optimization Hub,
  - [ ] Resource tagging (via Tag Editor)
  - [ ] AWS Trusted Advisor
- [ ] Establish:
  - [ ] Cost visibility dashboards
  - [ ] Monthly cost budget reports (Saving it in S3)

H. Working with the Channel Partner with better support (for all the heavy-lifting, double checking and technical consultation),

- [ ] Connect with a partner for AWS Support,
- [ ] Propose and have a consultation on the prepared document - Observation, Optimization and Modernization,
- [ ] Evaluate the proposal with a consultant from AWS and the dedicated partner,
- [ ] Need Support on
  - [ ] Cloud Billing Support (Discounts and credits if available)
  - [ ] Dedicated Support engineer from the partner - on demand
  - [ ] AWS Solutions Architect (with EKS Expertise) from AWS for high ticket supports
  - [ ] Support on Partner programs
- [ ] Trusted Advisors support if necessary
- [ ] Technical Support Checklist
  - [ ] Resource Discovery
  - [ ] Removal of stale and optimizing under-utilized workloads
  - [ ] Cost Optimization Support and Consultation

- [ ] Lift and Shift Workloads Migration support (for prod workloads)
- [ ] CPU Migration (x86 to Graviton ARM instances)
  - [ ] Evaluation
    - [ ] Performance
    - [ ] Cost
  - [ ] Benchmarking
    - [ ] Performance
    - [ ] Latency
- [ ] VPN Setup (Hybrid Environment - On-Prem and Cloud)
  - [ ] Establish seamless connection between AWS and On-Prem Endpoints,
  - [ ] Site-to-site tunneling between the organization and DC Endpoints,
  - [ ] Wireguard
- [ ] Modernization - Help us optimize, unify the platform, replace AWS Managed Billable Resource Services with the Right alternate OSS Selfhosted Tools.
- [ ] Consult Right alternate self-hosted tools on the organization's On-prem DataCentre to all the AWS managed services.
- [ ] Help us to keep up with the AWS Well-Architected Framework
- [ ] Evaluate our infra to comply and consult to follow best practices - complied to the AWS Well Architected Framework.
- [ ] Consultation on safest approach to perform each of the above planned modification.

First Meet with Ilios Digital - Meeting Agenda:

1. Introduction
2. Partner Introduction
3. Partner Capabilities
4. AWS Powered Capabilities
5. Pulse's Pain Points (scattered resources, reorganizing the infra, IaC Practices, Config Management)
6. with which and all they can able to assist with us, (Well Arch Framework, Cost Optimization and more)
7. AWS Direct support as Trusted Advisors
8. Moving Forward- mail: rithish (devops) rithishsamsdp@gmail.com, Kathir: Hiroshan Gopallapillai (chairman): hiroshan@pulse.in

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